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Chemical constituents from *Hericium erinaceus* and their ability to stimulate NGF-mediated neurite outgrowth on PC12 cells

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ABSTRACT

One new meroterpenoid, named hericenone K (**11**), along with 10 known compounds (**1**–**10**), ergosterol peroxide (**1**), cerevisterol (**2**), 3β,5α,9α-trihydroxy-ergosta-7,22-dien-6-one (**3**), inoterpene A (**4**), astradoric acid C (**5**), betulin (**6**), oleanolic acid (**7**), ursolic acid (**8**), hemisceramide (**9**), and 3,4-dihydro-5-methoxy-2-methyl-2-(4′-methyl-2′-oxo-3′-pentenyl)-9(7*H*)-oxo-2*H*-furo[3,4-*h*]benzopyran (**10**), was isolated from the fruiting bodies of the mushroom *Hericium erinaceus*. Their structures were characterized on the basis of spectroscopic methods, as well as through comparison with previously reported data. Compounds **3**–**6**, **8**, and **9** were isolated from *Hericium* species for the first time. Compounds **10** and **11** was suggested to be racemic by the CD spectrum data and specific rotations, which ware resolved by chiral HPLC into respective enantiomers. Compounds **1**–**3**, (±)-**10**, (−)-**10** and (+)-**10** in the presence of NGF (20 ng/mL) exerted a significant increase in neurite-bearing cells.

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Nerve growth factor (NGF) is the first and best characterized neurotrophic factor, that is, recognized as an important regulatory substance in the nervous system.[1] Thus, NGF is expected to have therapeutic efficacy for the treatment of neurodegenerative disease such as Alzheimer's disease and Parkinson's disease. However, it cannot cross the brain blood barrier because it is a high-molecular polypeptide and is easily metabolized by peptidases under physiological conditions.[1] To address this issue, considerable efforts have been made to find small molecules that have neurotrophic properties and/or that are capable of enhancing the action of endogenous neurotrophic factors.

Hericium erinaceus (Bull.: Fr) Pers. is a traditional edible mushroom belonging to the family Hericiaceae and has been used as an herbal medicine in East Asian countries. It is called Yamabushitake in Japanese, Houtougu (monkey head) in Chinese, and Lion's Mane in English after its shape.[2] In China, the fruiting bodies of *H. erinaceus* have been used for the treatment of dyspepsia, gastric ulcers, and enervation.[3] The chemical components of *H. erinaceus* include two specific classes of natural compounds, hericenones (meroterpenoids) and erinacines (cyathane diterpenoids), and their biological activities are well known. Hericenones and erinacines can induce the biosynthesis of NGF in nerve cells in vitro, such as hericenones C–E[4] and erinacines A–I,[5] isolated from the mycelia of *H. erinaceus*. In addition, some aromatic compounds, hericenone A, isohericenone and isohericerin exhibited antiproliferative activity against various cancer cell lines.[6,7]

In our ongoing effort to discover biologically active secondary metabolites from higher Basidiomycetes,[8–18] chemical investigation of the EtOAc fraction from the fruiting bodies of *H. erinaceus*[19] has led to the isolation of one new aromatic compound (**11**), together with 10 known compounds, ergosterol peroxide (**1**), cerevisterol (**2**), 3β,5α,9α-trihydroxy-ergosta-7,22-dien-6-one (**3**), inoterpene A (**4**), astradoric acid C (**5**), betulin (**6**), oleanolic acid (**7**), ursolic acid (**8**), hemisceramide (**9**), and 3,4-dihydro-5-methoxy-2-methyl-2-(4′-methyl-2′-oxo-3′-pentenyl)-9(7*H*)-oxo-2*H*-furo[3,4-*h*]benzopyran (**10**). In this Letter, we describe the isolation, structural elucidation, and neurotrophic activity of some isolates.

The EtOAc-soluble portion of an ethanol extract of the dried fruiting bodies of *H. erinaceus* was subjected to a series of column chromatographies over silica gel, Sephadex LH-20, RP-18, and semi-preparative reverse-phase HPLC. The purification resulted in the isolation of 11 compounds (**1**–**11**), including one new aromatic compound (**11**) and ten known compounds,[20] namely, ergosterol peroxide (**1**), cerevisterol (**2**),[21] 3β,5α,9α-trihydroxy-ergosta-7,22-dien-6-one (**3**),[22] inoterpene A (**4**),[23] astradoric acid C (**5**),[24] betulin (**6**),[25] oleanolic acid (**7**),[26] ursolic acid (**8**),[27] hemisceramide (**9**),[28]

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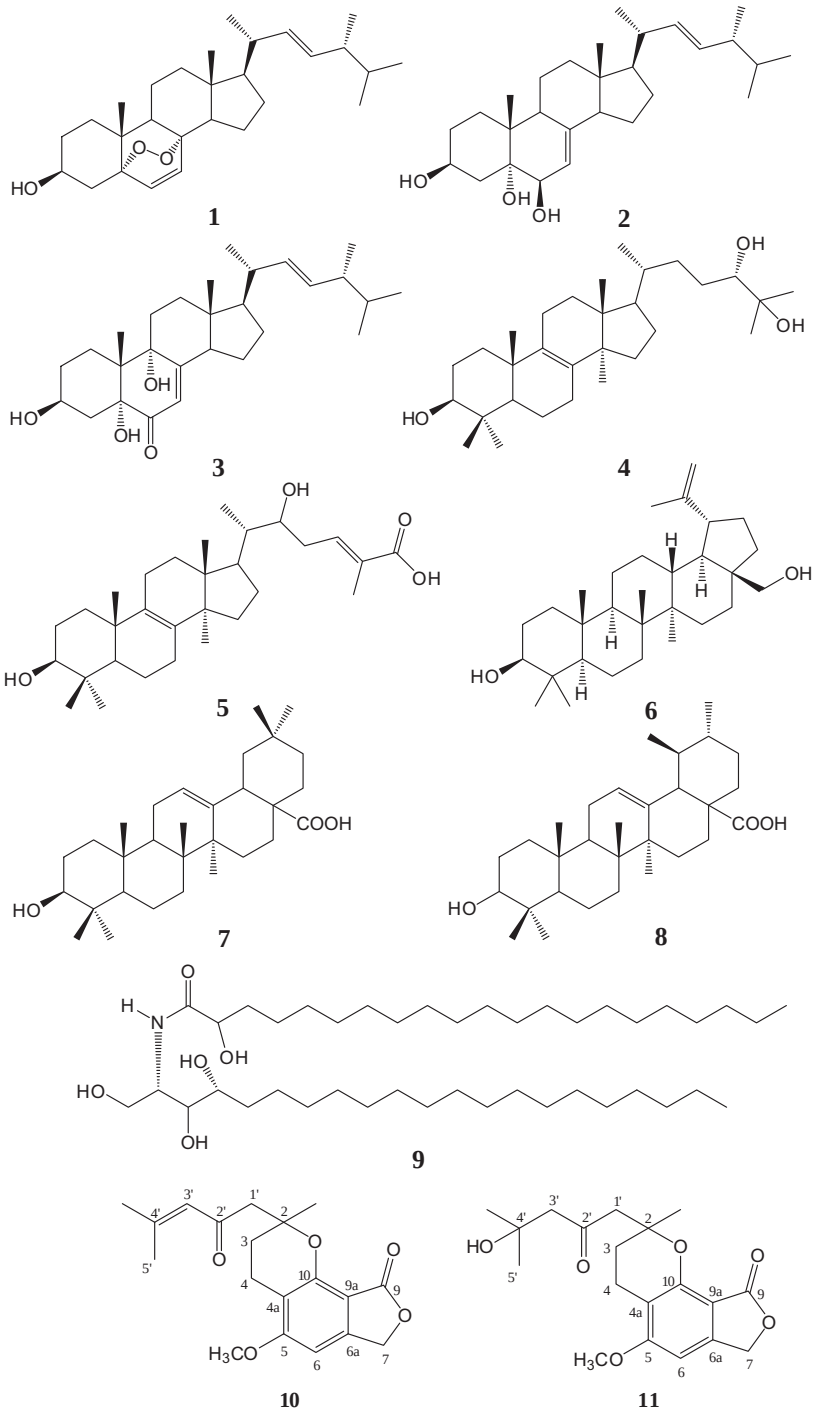

Figure 1. Structures of compounds **1**–**11**.

3,4-dihydro-5-methoxy-2-methyl-2-(4′-methyl-2′-oxo-3′-pentenyl)-9(7*H*)-oxo-2*H*-furo[3,4-*h*]benzopyran (**10**)[29] (Fig. 1). Among them, eight compounds (**3**–**6**, **8** and **9**) were reported from *Hericium* species for the first time.

The compound **10**[30] was characterized by comparing its physical constants and spectral data with those reported in the literature.[29] The complete assignment of all the protons and carbon atoms was accomplished by DEPT, HSQC and HMBC experiments. Despite the presence of an asymmetric carbon at C-2, the specific rotation of **10** was almost zero, suggesting it was racemic. This was proved by chiral reverse-phase HPLC,[31] in which **10** was separated into two peaks [peak A, (−)-**10**: $[\alpha]_D^{21}$ −57.5° (*c* 0.12, MeOH); peak B, (+)-**10**: $[\alpha]_D^{21}$ +56.2° (*c* 0.13, MeOH)].[32] The computational electronic circular dichroism (ECD) spectra of (+)-**10** and (−)-**10** exhibited mirror image CD curves [(+)-**10**: 260 nm ($\Delta\varepsilon$ + 14.3), 241 nm ($\Delta\varepsilon$ − 16.7); (−)-**10**: 260 nm ($\Delta\varepsilon$ − 15.3), 240 nm ($\Delta\varepsilon$ + 19.6)], confirming their enantiomeric relationship (Fig. 2). It should be worth mentioning that compound **10** was isolated earlier as a racemic mixture and its absolute configuration has not been studied yet.[7,29]

To determine the absolute configuration of a pair of enantiomers, (−)-**10** and (+)-**10**, a comparison was made between the

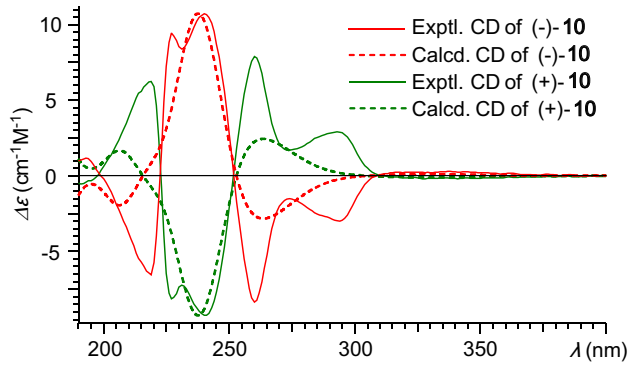

Figure 2. Experimental and theoretical ECD spectra of (−)-**10** and (+)-**10** in MeCN.

experimental and calculated CD spectra by the time-dependent density functional theory (TDDFT) method.[33] Since the compound **10** possesses only one chiral center, the ECD curves of *R* and *S* enantiomers are theoretically symmetric around zero-line in 2D coordinate graph. Thus, the *S*-enantiomer was arbitrarily selected to calculate. Its conformers were searched under MMFF94s force field in Conflex 6.7 and further optimized by B3LYP/6–31+G(d,p) with CPCM solvent model in Gaussian 09. The four lowest-energy conformers (within 3 cal/mol cutoff) were then calculated their ECD by TDDFT method (cam-B3LYP/TZVP). The final ECD curve of *S*-enantiomer was surmised according to the Boltzmann distributions of the stable conformers. As a result, in Figures 2 and 3, the ECD of the lowest-energy conformer *S*-**10** possesses same Cotton effect with that of (−)-**10** and its absolute configuration was assigned therein. The experimental ECD curve of (+)-**10** possesses absolutely antipodal Cotton Effects to that of experimental and calculated of *S*-enantiomer, therefore, its absolute configuration was assigned as *R*.

Compound **11**[34] was obtained as colorless oil. Its molecular formula was determined as $C_{19}H_{24}O_6$ by HR-ESI-MS *m/z* 371.1464 $[M+Na]^+$ (calcd for $C_{19}H_{24}O_6$, 371.1471). The IR spectrum shows the presence of a hydroxyl group (3436 cm^{-1}), a γ-lactone (1748 cm^{-1}), a ketone group (1697 cm^{-1}) and a benzene ring (1608 cm^{-1}), and the UV spectrum reveals the presence of an aromatic moiety (260 nm). The ^1H NMR spectrum contained peaks consistent with an aromatic proton at δ_H 6.70 (s, H-6), a methoxy group at 3.93 (s), five methylenes at δ_H 1.93 (m, H-3a), 2.02 (m, H-3b), 2.63 (m, H-4a), 2.68 (m, H-4b), 2.84 (d, *J* = 14.6 Hz, H-1′a), 2.93 (d, *J* = 14.6 Hz, H-1′b), 2.78 (d, *J* = 15.9 Hz, H-3′a), 2.89 (d, *J* = 15.9 Hz, H-3′b) and 5.21 (m, H-7), and three methyls at δ_H 1.23 (s, 4′-Me), 1.25 (s, H-5′), 1.39 (s, 2-Me). The ^{13}C NMR and DEPT spectra showed 19 carbon signals corresponding to a methoxy carbon at δ_C 56.8, six signals in the aromatic region of δ_C 96.7–165.3, five methylene and three methyl signals between 17.7 and 70.4, four quarternary carbons including two carbonyl ones [(δ_C 211.3 (C-2′), 171.8 (C-9)]. The chroman skeleton was determined by ^1H–^1H COSY correlations between H-3 and H-4 (Fig. 4). The position of the methoxy group was assigned by the HMBC correlation from the methoxy to C-5. The presence of the ketone (C-2′) was indicated by the characteristic chemical shift at δ_C 211.3. The γ-lactone moiety was identified by correlations between H-6/C-7, H-7/C-9, C-10 in the HMBC spectrum (Fig. 4). The side chain moiety attached to C-2 was elucidated by the HMBC correlations (H-1′/C-2′; H-3′/C-2′, C-4′-Me, C-5′; H-5′/C-3′, C-4′, C-4′-Me; C-4-Me/C-3′, C-4′, C-5′). The NMR data of **11** were very similar to those of **10**. Only difference is that the $\Delta^{3'}$ double bond of **10** was replaced by sigma bond between C-3′ (δ_C 56.8) and C-4′ (δ_C 70.9) in **11**. The complete assignment of all the protons and carbon atoms was accomplished by DEPT, HSQC and HMBC experiments as shown in Table 1. Thus, from the above data analysis, the structure of compound **11** was determined as shown, and named hericenone K.

Likewise, the specific rotation of (±)-**11** was almost zero, suggesting it was a pair of (−)- and (+)-antipodes. This was confirmed by chiral reverse-phase HPLC,[31] in which **11** was separated into

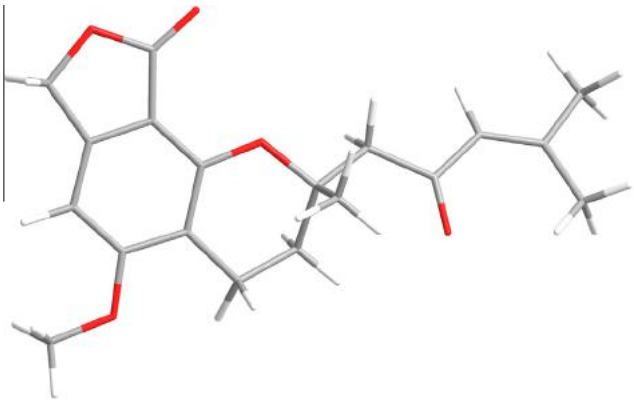

Figure 3. The structure of the lowest-energy conformer of *S*-**10**.

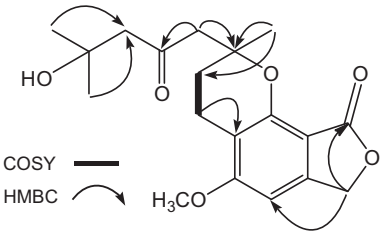

Figure 4. Key COSY and HMBC correlations of compound **11**.

Table 1
^1H (500 MHz) and ^{13}C NMR (125 MHz) spectroscopic data of compound **11** in CD_3OD

No.	δ_H (*J* in Hz)	δ_C	DEPT	HMBC
2		78.0	C	
3	1.93 (1H, m*) 2.02 (1H, m*)	31.0	CH_2	C-2, C-4, C-1′, C-4a, C-2-CH_3
4	2.63 (1H, m*) 2.68 (1H, m*)	17.7	CH_2	C-2, C-3, C-5, C-10, C-4a
4a		111.7	C	
5		165.3	C	
6	6.70 (1H, s)	96.7	CH	C-5, C-7,C-10, C-4a, C-6a, C-9a
6a		151.5	C	
7	5.21 (2H, m)	70.4	CH_2	C-5, C-6, C-9, C-10, C-4a, C-6a, C-9a
9		171.8	C	
9a		106.6	C	
10		154.0	C	
2-CH_3	1.39 (3H, s)	24.0	CH_3	C-2, C-3, C-1′
5-OCH_3	3.93 (3H, s)	56.8	CH_3	C-5
1′	2.84 (1Ha, 14.6), 2.93 (1Hb, 14.6)	54.0	CH_2	C-2, C-3, C-2-CH_3
2′		211.3	C	
3′	2.78 (1Ha, d, 15.9), 2.89 (1Hb, d, 15.9)	56.8	CH_2	C-4′, C-5′, C-4′-CH_3
4′	7.90	70.9	C	
5′	1.25 (3H, s)	29.4	CH_3	C-3′, C-4′, C-4′-CH_3
4′-CH_3	1.23 (3H, s)	29.6	CH_3	C-3′, C-4′, C-5′

* Overlapped due to a mixture of (−)-11 and (+)-11.

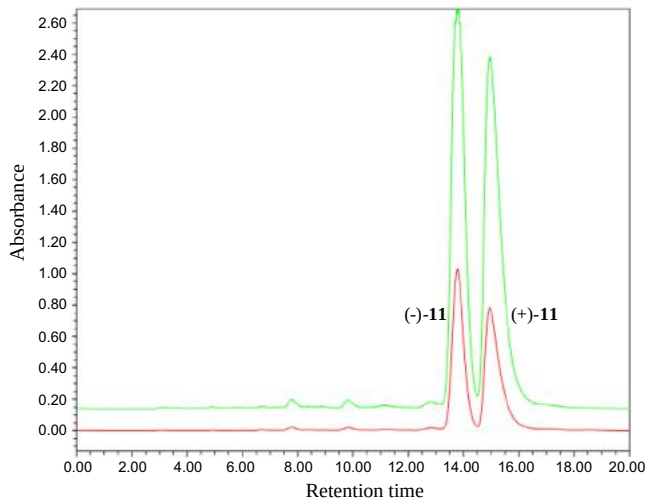

Figure 5. Chiral HPLC separation chromatogram of **11** on chiral Daicel Chiralpack AD-H column (250 × 4.6 mm, *n*-hexane–2-propanol = 8:2, 1 mL/min, red line: detection at 254 nm, green line: detection at 220 nm).

two enantiomers, (−)-**11** and (+)-**11**, in a ratio of almost 1:1 (Fig. 5), but its pure enantiomers were not obtained due to limited amounts.

Lai et al.[35] have shown *H. erinaceus* aqueous extract in combination with exogenous NGF on the neurite outgrowth stimulation of neuroblastoma-glioma cell NG108-15. Moreover, hericenones C–E from *H. erinaceus* have been reported to be effective stimulators of NGF synthesis.[4,36] There was, however, no report on the biological activity of the known compound **10**.

Cell viability was evaluated by the SRB reduction assay[37] and the neurite outgrowth (neuritogenic activity) of compounds **1**–**9**, (±)-**10**, (−)-**10** and (+)-**10** were tested using rat pheochromocytoma (PC-12) cells as a model system of neuronal differentiation.[18] Cell viability assays showed that the test compounds had no toxic effects on PC-12 cells at concentrations of 10 and 20 μM. To study whether these compounds exhibit neurotrophic properties in PC-12 cells, neurite outgrowth in the culture media was measured.[38] The cells were incubated with compounds **1**–**10** at concentrations of 10 and 20 μM in the absence and presence of NGF (20 ng/mL). NGF was used as a positive control. Only compounds **1**, **2**, **3**, (±)-**10**, (−)-**10** and (+)-**10** in combination with NGF (20 ng/mL) exerted a significant increase in neurite-bearing cells compared to NGF-treated PC-12 cells (Fig. 6). The percentage of neurite-bearing cells for cells treated with 20 μM (±)-**10**, (−)-**10** and (+)-**10** in the presence of NGF (20 ng/mL) reached 19.96 ± 0.32%, 17.85 ± 0.36%, and 21.79 ± 0.49%, respectively (Fig. 6). These were compatible with those treated with 20 ng/mL NGF (15.32 ± 0.25%) and significantly higher than those of negative control (2.15 ± 0.23%). In contrast, the test compounds alone had no effect on neurite outgrowth in PC-12 cells in the absence of NGF (20 ng/mL).

In conclusion, one new compound **11**, along with 10 known compounds (**1**–**10**) was isolated from the fruiting bodies of *H. erinaceum*, and their structures were elucidated by spectroscopic analyses. The chiral HPLC was used to separate the (−)- and (+)-antipodes of the two enantiomers, **10** and **11**. Compounds (±)-**10**, (−)-**10** and (+)-**10** exhibited high neurite outgrowth-promoting activity in NGF-induced PC12 cells, while this activity of compounds **1**, **2** and **3** was weak. These data suggested that the aromatic compounds are further neuroactive substances found in *H. erinaceum*.

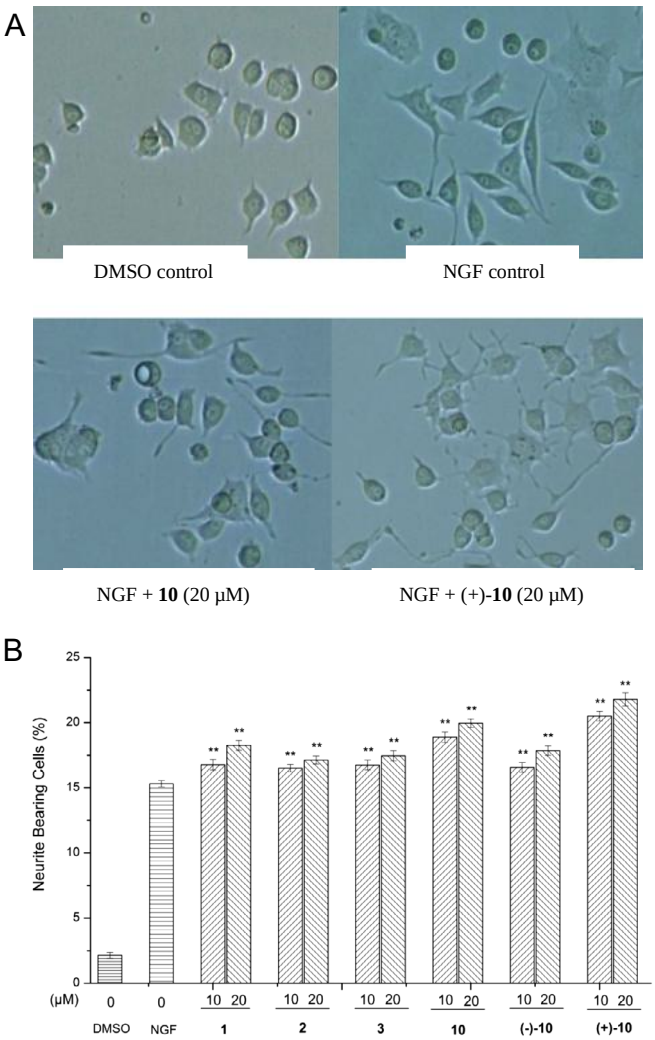

Figure 6. Effects of compounds **1**, **2**, **3**, (±)-**10**, (−)-**10** and (+)-**10** on the neurite outgrowth in NGF-induced PC-12 cells. PC12 cells were seeded in poly-L-lysine-coated 24-well plates in normal serum medium for 24 h, then shifted to low serum medium (1% HS and 0.5% FBS) and exposure to vehicle (0.1% DMSO) as a negative control, NGF (20 ng/mL) as a positive control, or NGF (20 ng/mL) + **1**, **2**, **3**, (±)-**10**, (−)-**10** and (+)-**10** (10–20 μM) for an additional 48 h. (A) Cell morphology was observed and photographed as described in Measurement of neurite outgrowth.[38] (B) Neurite bearing cells were analyzed as described in Measurement of neurite outgrowth.[38] Data represent the mean ± SD of three dependent experiments. $^{**}p$ <0.01 represent significant differences compared with NGF-treated PC-12 cells (positive control) (ANOVA followed by Dunnett's *posthoc* multiple comparisons test).

Acknowledgements

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19. Dried fruiting bodies of *H. erinaceus* were purchased from QingYuan Green & Best Food Co., Ltd, Zhejiang Province, China in October 2013. A voucher specimen was deposited at the College of Science, Northwest A&F University.
20. *Extraction and isolation:* Dried fruiting bodies (20 kg) of *H. erinaceus* were extracted with 95% ethanol under reflux. The ethanol extract (5.7 kg) was suspended in water and partitioned with EtOAc. The EtOAc extract (280 g) was subjected to silica gel column chromatography (CC) with a gradient of petroleum ether (PE)–acetone–MeOH (10:0:0, 9:1:0, 7:3:0, 5:5:0, 0:0:1) to give 9 fractions (Fr.1A–1I). The fraction 1E (9.5 g) was separated using silica gel CC with a gradient of PE–EtOAc (8:1, 3:1) to give 5 subfractions (Fr.1E-1–1E-5). Fraction 1E-3 was subjected to silica gel CC with PE–EtOAc (9:1) to give **6** (40 mg). The fraction 1E-5 was purified by RP-18 CC (MeOH–H_2O, 10–100%) to obtain **1** (385 mg). The fraction 1F (17.9 g) was subjected to silica gel CC with a gradient of $CHCl_3$–acetone (100:1, 10:1) to give 5 fractions (Fr.1F-1–1F-5). Fraction 1F-2 was separated using RP-18 CC with a MeOH–H_2O elution solvent to give 5 fractions (Fr.1F-2-1–1F-2-5). Fraction 1F-2-4 was separated by Sephadex LH-20 ($CHCl_3$–MeOH = 1:1) and further purified by PTLC ($CHCl_3$–acetone, 25:1) to give **5** (9.4 mg). Fraction 1F-4 was submitted to silica gel CC with a gradient of $CHCl_3$–acetone (50:1, 20:1, 10:1, 5:1, 1:1) to give 5 fractions (Fr.1F-4-1–1F-4-5). Fraction 1F-4-1 was fractionated using silica gel CC with PE–acetone (4:1, 2:1) to give 3 fractions (Fr.1F-4-1-1–1F-4-1-3). Fraction 1F-4-1-2 was fractionated by LH-20 ($CHCl_3$–MeOH = 1:1) and RP-18 (MeOH–H_2O, 10–100%) and further purified by silica gel CC ($CHCl_3$–acetone, 40:1) to give **7** (6.3 mg) and **8** (4.5 mg). Fraction 1F-4-1-3 was separated by LH-20 ($CHCl_3$–MeOH = 1:1) and RP-18 (MeOH–H_2O, 10–100%) and further purified by silica gel CC ($CHCl_3$–MeOH, 50:1) to provide **4** (2.1 mg). Fraction 1G (42 g) was separated using silica gel CC with a gradient of PE–acetone (8:1, 4:1, 2:1, 1:1, 1:2) to give 5 fractions (Fr.1G-1–1G-5). Fraction 1G-2 was subjected to RP-18 CC with MeOH–H_2O to obtain 7 fractions (Fr. 1G-2-1–1G-2-7). Fraction 1G-2-2 was separated by LH-20 ($CHCl_3$–MeOH = 1:1) and further purified by PTLC ($CHCl_3$–MeOH, 50:1) to give **11** (5 mg). Fraction 1G-2-3 was fractionated using LH-20 ($CHCl_3$–MeOH = 1:1) and PTLC ($CHCl_3$–acetone, 30:1) and then purified using HPLC (70% MeOH in H_2O, 2 mL/min) to yield **10** (t_R = 7.7 min, 42.6 mg). Fraction 1G-3 was subjected to silica gel CC with $CHCl_3$-acetone (20:1, 10:1) and LH-20 ($CHCl_3$–MeOH = 1:1) and then purified by PTLC ($CHCl_3$–acetone, 25:1) to give **9** (4 mg). Fraction 1H (42 g) was subjected to silica gel CC with a gradient of PE–acetone (6:1, 3:1, 2:1, 1:1, 1:2) to give 5 fractions (Fr.1H-1–1H-5). Fraction 1H-3 was separated by silica gel CC with a gradient of $CHCl_3$–acetone (50:1, 20:1, 10:1, 5:1, 1:1) to give 5 fractions (Fr.1H-3-1–1H-3-5). Fraction 1H-3-3 was subjected to silica gel CC with $CHCl_3$–MeOH (50:1, 20:1, 10:1) and RP-18 (MeOH–H_2O, 10–100%) and further purified by silica gel CC ($CHCl_3$–MeOH, 20:1) to give **2** (420 mg). Fraction 1H-3-5 was subjected to silica gel CC with a gradient of PE–acetone (6:1, 5:1, 4:1), LH-20 ($CHCl_3$–MeOH = 1:1) and then purified by PTLC (PE–acetone, 4:1) to afford **3** (30 mg).
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30. *Compound* **10**: colorless oil; $[\alpha]_D^{21}$ = ±0 (*c* 0.14, MeOH); ESI MS (positive) *m*/*z* 331.12 $[M+H]^+$, 353.26 $[M+Na]^+$.
31. *Chiral resolution:* Chiral separation was performed on a CHIRALPAK AD-H column (5 μm, 4.6 mm i.d. × 250 mm, Chiral Technologies (China) CO., Ltd). Compound **10** was dissolved in 2-propanol, and then purified using chiral HPLC (*n*-hexane–2-propanol = 8:2, 1 mL/min) to give compound (−)-**10** (t_R = 13.0 min, 8.3 mg) and (+)-**10** (t_R = 14.9 min, 3.7 mg). Compound **11** was dissolved in 2-propanol, and then analyzed using HPLC (*n*-hexane–2-propanol = 8:2, 1 mL/min) to display compound (−)-**11** (t_R = 13.8 min) and (+)-**11** (t_R = 15.0 min).
32. (*S*)-(−)-**10**: colorless oil; $[\alpha]_D^{21}$ −57.5 (*c* 0.12, MeOH). CDλ (*c* = 0.367 mg/mL, MeCN) (nm) (Δε): 294 (−5.5), 260 (−15.3), 240 (+19.6), 227 (+17.3), 219 (−12.0), 193 (+2.1). ESIMS (positive) *m*/*z* 353.24 $[M+Na]^+$. (*R*)-(+)-**10**: colorless oil; $[\alpha]_D^{21}$ +56.2 (*c* 0.13, MeOH). CDλ (*c* = 0.363 mg/mL, MeCN) (nm) (Δε): 293 (+5.3), 260 (+14.3), 241 (−16.7), 227 (−14.7), 219 (+11.3). ESIMS (positive) *m*/*z* 353.26 $[M+Na]^+$.
33. *Computational section*: A preliminary conformational search was performed in Conflex6.7 using MMFF94s forcefield, according to previously reported methods. Conformers were saved and further optimized using the density functional theory (DFT) method and CPCM solvent model at B3LYP/6-31+G(d, p) level in Gaussian 09 software package. Frequency was calculated at the same level of theory to check optimized results. The four low-energy conformers with populations greater than 1% and without imaginary frequencies were submitted to ECD calculation by the TDDFT (cam-B3LYP/TZVP) method associated with CPCM solvent model in MeCN. The excitation energies (*E*), oscillator strength (*f*), rotatory strength in velocity form (R_{vel}), and rotatory strength in length form (R_{len}) of the lowest 32 excited states were calculated. ECD spectra of different conformers were summated in SpecDis 1.62 according to their Boltzmann-calculated distributions. The summated curve was adjusted +5 nm UV-shift and 0.3 ev.
34. *Hericenone K* **11**: colorless oil; $[\alpha]_D^{21}$ = ±0 (*c* 0.12, MeOH); IR ν_{max}, 3436, 1748, 1697, 1608 cm^{-1}; UV(MeOH) λ_{max} (logε) 260 nm (4.23); ^1H (500 MHz, CD_3OD) and ^{13}C NMR data (125 MHz, CD_3OD), see Table 1; ESIMS (positive) *m*/*z* 371.04 $[M+Na]^+$; HR ESIMS *m*/*z* 371.1464 $[M+Na]^+$ (calcd for $C_{19}H_{24}O_6Na$, 371.1471).
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38. Measurement of neurite outgrowth in PC-12 cells: Morphological analysis and quantification of neurite-bearing cells were performed as described previously. PC-12 cells were seeded on poly-L-lysine-coated 24-well plates at a density of 2 × 104 cells/mL in normal serum medium for 24 h. The F-12 medium containing low serum (1% HS and 0.5% FBS) was replaced prior to exposure to vehicle (0.1% DMSO) or indicated reagents. The cells were treated with tested compounds at various concentrations (10–20 μM) in the absence or presence of NGF (20 ng/mL). Cells without treatment served as a negative control. Cells treated with NGF (20 ng/mL) served as a positive control. One concentration experiment was repeated in three wells. After incubation for an additional 48 h, neurite outgrowth of PC12 cells was observed under an inverted microscope using phase-contrast objectives and photographed by the digital camera. Five images were selected randomly under a microscope for each well. At least 100 cells in each randomly separated field were scored, and cells with neurites greater than or equal to the length of one cell body were positive for neurite outgrowth and expressed as a percentage of the total cell number in five fields.